

Quantum Entanglement and Quantum Computing

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Qubit: A bit of data represented by either a $|0\rangle$ or a $|1\rangle$.

Qubits can be forced into superposition by adding their state vectors

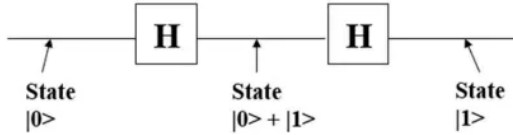
$$|\psi\rangle = \alpha_0 |0\rangle + \alpha_1 |1\rangle$$

where $\alpha_i \in \mathbb{C}$ are probability amplitudes with $\sum_i |\alpha_i|^2 = 1$. It is important to remember that **Qubits are always in superposition.** and that an n-Qubit register can represent 2^{n-1} values. **Entanglement:** Correlations within a superposition.

Quantum Gates

A kind of a gate where the inputs can be derived from the outputs. In other words, a Quantum gate is **Reversible**.

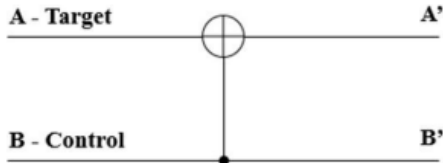
Hadamard Gate



Above is a representation of a Hadamard gate, it involves one Qubit and puts that certain qubit into a superposition. Mathematically it can be represented as

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}; H|\psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} a+b \\ a-b \end{bmatrix}$$

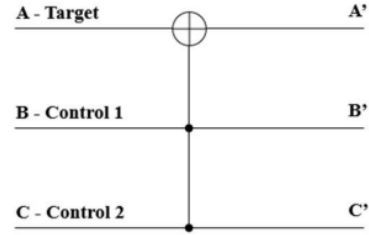
Controlled Not Gate



A	B	A'	B'
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	1

Operates on Two Qubits, if the bit on the control line is 1, Inverts A.

Controlled Controlled Not Gate



Also known as a Toffoli gate, the CCN gate inverts the Target bit if both of the Control bits are 1. This operates on 3 Qubits rather than 2. the truth table looks as such:

A	B	C	A'	B'	C'
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	1	1
1	0	0	1	0	0
1	0	1	1	0	1
1	1	0	1	1	1
1	1	1	1	1	0

Shor's Algorithm

Proves that a Quantum Computer is capable of factoring very large numbers in polynomial time.

Step 1: Classical preprocessing - check if N is even/prime/perfect power. Choose random $a < N$ with $\gcd(a, N) = 1$.

Step 2: Quantum period finding - find period r of $f(x) = a^x \bmod N$ using QFT:

$$QFT|j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i j k / N} |k\rangle$$

Step 3: Classical post-processing - extract r from measurement. Check if r is even and $a^{r/2} \not\equiv -1 \pmod{N}$.

Step 4: Factor extraction - compute $\gcd(a^{r/2} \pm 1, N)$ to get non-trivial factors.

Stern-Gerlach Experiment

Demonstrates quantum spin and the quantization of angular momentum in atoms.

Setup: Silver atoms beam passes through non-uniform magnetic field with gradient $\frac{\partial B_z}{\partial z}$.

Classical Expectation: Continuous distribution of deflections based on random orientation of magnetic dipoles.

Quantum Result: Beam splits into exactly two discrete spots, corresponding to spin states $|\uparrow\rangle$ and $|\downarrow\rangle$.

Force Equation: $F_z = \mu_z \frac{\partial B_z}{\partial z}$ where $\mu_z = \pm \frac{\mu_B}{2}$

Significance: Proves intrinsic angular momentum (spin) is quantized with $s = \frac{1}{2}$ for electrons, establishing foundation of quantum mechanics.

Dirac Bra-Ket Notation

Mathematical formalism for describing quantum states and operations in Hilbert space.

State Representation: Ket $|\psi\rangle$ represents quantum state, bra $\langle\phi|$ represents complex conjugate. Complete state: $\langle\phi|\psi\rangle$.

Inner Product: $\langle\phi|\psi\rangle = \int_{-\infty}^{\infty} \phi^*(x)\psi(x)dx$ gives probability amplitude for transition from $|\psi\rangle$ to $|\phi\rangle$.

Projection: $|\phi\rangle\langle\phi|$ projects onto state $|\phi\rangle$. Action on state: $|\phi\rangle\langle\phi|\psi\rangle$.

Position Representation: $\langle x|\psi\rangle = \psi(x)$ gives wavefunction in position basis.

Probability Density: $|\psi(x)|^2 = |\langle x|\psi\rangle|^2$ represents probability density of finding particle at position x .

Normalization: $\langle\psi|\psi\rangle = \int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1$

Note: Refer to Quantum Math PDF for additional mathematical foundations and detailed derivations.

Bloch Sphere

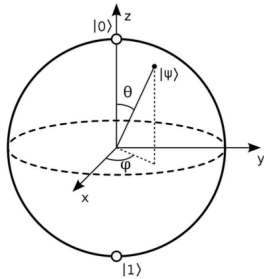


Figure 1: Bloch sphere representation showing polar angle θ and azimuthal angle ϕ

The Bloch sphere provides a geometric visualization of qubit states. Any qubit state can be represented as a point on the unit sphere using spherical coordinates:

$$|\psi\rangle = \cos(\theta/2) |0\rangle + e^{i\phi} \sin(\theta/2) |1\rangle$$

where $\theta \in [0, \pi]$ is the polar angle and $\phi \in [0, 2\pi]$ is the azimuthal angle. The north pole corresponds to $|0\rangle$, south pole to $|1\rangle$, and the equator represents equal superposition states.

Qubits and States

A qubit is the fundamental unit of quantum information, existing as a superposition of basis states $|0\rangle$ and $|1\rangle$:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

where $\alpha, \beta \in \mathbb{C}$ are complex amplitudes satisfying the normalization condition:

$$|\alpha|^2 + |\beta|^2 = 1$$

The probability of measuring the qubit in state $|0\rangle$ is $|\alpha|^2$ and in state $|1\rangle$ is $|\beta|^2$. Upon measurement, the state collapses to the measured eigenstate.

Density Matrix Formalism

Density Operator, Pure and Mixed States

The density operator provides a complete statistical description of quantum systems:

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

where p_i are classical probabilities and $\sum_i p_i = 1$.

Pure States: When the system is in a definite quantum state:

$$\rho_{\text{pure}} = |\psi\rangle\langle\psi|$$

Characteristic property: $\rho^2 = \rho$ and $\text{Tr}(\rho^2) = 1$

Mixed States: Statistical mixture of quantum states:

$$\rho_{\text{mixed}} = \frac{1}{2} |0\rangle\langle 0| + \frac{1}{2} |1\rangle\langle 1|$$

Characteristic property: $\rho^2 \neq \rho$ and $\text{Tr}(\rho^2) < 1$

Traces of Density Matrix

The trace operation is fundamental in quantum mechanics:

$$\text{Tr}(\rho) = 1$$

For composite systems, the **partial trace** allows us to obtain the reduced density matrix:

$$\rho_B = \text{Tr}_A(\rho_{AB}) = \sum_i \langle i|_A \rho_{AB} |i\rangle_A$$

This operation "traces out" subsystem A to focus on subsystem B. The partial trace is essential for studying entanglement and separability.

Quantum Superposition and Operations

Superposition of Quantum States

Quantum superposition is the fundamental principle allowing qubits to exist in multiple states simultaneously:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Unlike classical probability, quantum amplitudes can interfere:

- **Constructive interference:** Amplitudes add coherently
- **Destructive interference:** Amplitudes cancel out

This enables quantum parallelism, where quantum algorithms can process multiple possibilities simultaneously.

Quantum gates are unitary operators that manipulate qubit states:

Single-qubit gates:

Pauli Gates:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Hadamard Gate:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Creates equal superposition: $H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

Phase Gates:

$$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}, \quad T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$$

Multi-qubit gates:

CNOT Gate:

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

The CNOT gate flips the target qubit if the control qubit is in state $|1\rangle$, creating entanglement between qubits.

Controlled-Z Gate:

$$\text{CZ} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$